# Factsheet: Simple random sampling

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**Summary**

This factsheet summarises simple random sampling {srs} and defines its expectation and variance.

*Before reading this factsheet, it is recommended that you read* [Guide: Introduction to sampling]

## Simple random sampling {srs; $\bar{y}$}

In simple random sampling, everything in the population has **equal probability** of being chosen. Also, every possible sample has **equal probability** of being chosen.

> **i** Definition: Probability of selecting a particular sample under srs
>
> If a population has $N$ individuals and you take a sample size of $n$, then:
> $P$(Selecting a particular sample) is
>
> $$\frac{1}{\binom{N}{n}}$$
>
> or
>
> $$\binom{n}{N}$$
>
> where:
>
> $$\binom{n}{N} = \frac{n!}{N!(n-N)!}$$

In srs, $\bar{y}$ is the sample mean of $y$, such that $\bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i$ and it is an unbiased estimator for the population mean $\mu$.

This table contains key properties of $\bar{y}$ for srs:

| | |
|---|---|
| Expectation $=$ E[$\bar{y}$] | $\mu$ |

| | |
|---|---|
| Variance = Var[$\bar{y}$] | $(1-f)\frac{S^2}{n}$ |
| Confidence interval (CI) | $\bar{y} \pm Z_{\frac{\alpha}{2}} \frac{s}{\sqrt{n}} \sqrt{(1-f)}$ |

where,

- $S^2$ = Finite population variance of y = $\frac{1}{(N-1)} \sum_{i=1}^{n} (y_i - \mu)^2$
- $f$ = Sampling fraction = $\frac{n}{N}$

# Further reading

# Version history

v1.0: initial version created 12/06 by Millie Harris as part of a University of St Andrews VIP project.

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